

Hyunin Lee

Email : hyunin@berkeley.edu | [linkedin](#) | [github](#) | [homepage](#)

Education

University of California, Berkeley

Ph.D. in Mechanical Engineering / Advisor: Dr.Somayeh Sojoudi

CA, United States

Aug. 2022 – May. 2027 (expected)

Seoul National University

B.S in Mechanical Engineering; summa cum laude

Seoul, Rep.of.Korea

Mar. 2015 – Feb. 2022

Research Statement

My research lies at the intersection of **reinforcement learning** and LLM-based multi-agent systems, focusing on non-stationarity, robustness, and currently **recursive self-improvement**. My work has appeared in ICML, ICLR, and NeurIPS, including oral presentations. I develop principled methods for building reliable self-improving multi-agentic systems that operate effectively under real-world uncertainty.

Publications / C: CONFERENCE, J: JOURNAL, P: PREPRINT

[P3] Recursive Harness Self-Improvement.

H. Lee, J. Xu, J. Seely, D. Lee, M. Zaharia and Y. Tang. *Preprint*. 2026. [pdf/blog]

[P2] Cross-attention Secretly Performs Orthogonal Alignment in Recommendation models.

H. Lee, Y. Zhang, H. Nguyen, X. Liu, N. Park, C. Jung, Y. Wang, Z. Wang, S. Sojoudi, X. Feng. *Under review*. 2026. [pdf/blog]

[P1] A Prospect Theoretic Rationality: Loss Aversion Can Accelerate Reinforcement Learning

H. Lee, C. Park, S. Sojoudi, N. Mehr. *preprint*. 2026. [pdf]

[C5] Position: AI Safety Must Embrace an Antifragile Perspective.

M. Jin, **H. Lee**. *ICML*. 2025. [pdf]

[C4] A Black Swan Hypothesis: The Role of Human Irrationality in AI Safety.

H. Lee, C. Park, D. Abel, M. Jin. *ICLR. 2025 & ICLR 2025 Advances in Financial AI Workshop* (Oral, Top 1.2%). [pdf]

[C3] Pausing Policy Learning in Non-stationary Reinforcement Learning.

H. Lee, M. Jin, J. Lavaei, and S. Sojoudi. *ICML*. 2024. (Oral, Top 1.2%) [pdf/codes/talk]

[J3] Policy-based Primal-Dual Methods for Concave CMDP with Variance Reduction.

D. Ying, M. Guo, **H. Lee**, Y. Ding, J. Lavaei, and Z. Shen. *JAIR*. 2025. [pdf /codes]

[C2] Tempo Adaptation in Non-stationary Reinforcement Learning.

H. Lee, Y. Ding, J. Lee, M. Jin, J. Lavaei, and S. Sojoudi. *NeurIPS*. 2023 [pdf/codes/slides]

[J2] Beyond Exact Gradients: Convergence of Stochastic Soft-Max Policy Gradient Methods with Entropy Regularization.

Y. Ding, J. Zhang, **H. Lee**, and J. Lavaei. *IEEE TAC*. 2025 [pdf]

[C1] Initial State Interventions for Deconfounded Imitation Learning.

S. Pfrommer, Y. Bai, **H. Lee**, and S. Sojoudi. *IEEE CDC*. 2023. [pdf]

[J1] Explainable Deep Learning Model for EMG Based Finger Angle Estimation Using Attention.

H. Lee, D. Kim, and Y. Park. *IEEE TNSRE*. vol. 30, pp. 1877-1886 2022. [pdf/codes]

Work Experience

SakanaAI

Research Intern

May 2026 –

Research Team

- Researching automating self-improving LLM-based multi-agent systems. (Intern Manager: Yujin Tang)

Meta

Research Scientist Intern

May 2025 – Oct 2025

Ranking AI Research Team

- Researched on transformer-based **multi-modal recommendation model**, enhancing ranking performance in sparse data setting. (Intern manager: Yong Zhang, Xue Feng)
- Proposed a novel cross-attention mechanism—orthogonal alignment—achieving 1.4x improvement in **scaling law efficiency**, validated on both Meta ranking model and public baselines. [p1]

OpenAI

Research Associate

Mar 2025 – Aug 2025

Human Data (Research) Team

- Engaged with the safety and preparedness team to evaluate an AI agent's ability to replicate ML research and to create a comprehensive rubric that defines objective success criteria for accurately reproducing given ML papers. (Manager: Benjamin Kinsella)

University of California, Berkeley

Graduate Student Researcher

Aug 2022 – Present

Advisor: Dr. Somayeh Sojoudi

- Research on **non-stationary reinforcement learning** and optimization under distributional shift. [p1, p2]
- Research on **Human-AI alignment** focusing on ensuring **safe AI systems** in the presence of unpredictable or non-standard human behaviour. [p1, p2, p3]
- Research on optimal workflow control of LLM-based Multi-agent.

OUTTA [Homepage] [LinkedIn]

Co-Founder

Aug 2021 – Present

Seoul, South Korea

- Deliver online AI courses to **800+** students annually across South Korea.

Knowledge AI

Machine Learning Engineer

Jul 2021 – Jul 2022

Boston, MA

- Implemented a **Bayesian inference algorithm** in Python to quantify students' mastery of mathematics topics.
- Developed a question-recommendation deep-learning system for an online math-learning platform, boosting student performance.

Seoul National University

Undergraduate Research Intern

Mar 2021 – Nov 2021

Soft Robotics & Bionics Lab

- Designed an **attention-based sequential decision-making algorithm** in Python to predict finger-joint angles from forearm EMG signals, improving accuracy by more than 10%.

Seoul National University

Undergraduate Research Intern

Sep. 2020 – Jun. 2021

Robot Learning Lab

- Develop **deep generative Q learning algorithm** to reconstruct a reward kernel using Python [pdf] [video]

Academic Service

Reviewer

ICLR 2024-2026, ICML 2024-2027, NeurIPS 2024-2027, RLC 2025, AAAI 2025-2026

Program Chair Committee

AAAI 2025 - 2026

Teaching Experience

Graduate Student Instructor, Statistics and Data Science for Engineers Spring, Fall 2025 / Spring 2026

Graduate Student Instructor, Dynamic Systems and Feedback Fall 2024

Teaching Tutor, Math and Physics for Freshman Spring 2019 / Spring 2020

Teaching Assistant, Dynamic Fall 2019

Teaching Assistant, Mechanical Product Design Fall 2020

Grants and Honors

Berkeley Summer Research Fellowship <i>Mechanical Engineering Department</i>	Summer 2024
NeurIPS scholar award <i>Conference on Neural Information Processing Systems</i>	Dec. 2023
Kwanjeong Abroad Scholarship <i>Kwanjeong Educational Foundation</i>	Fall 2022 – Present
Berkeley Fellowship for Graduate Study <i>Graduate Division</i>	Fall 2022 – Spring 2023
National Science & Technology Scholarship <i>Korea Student Aid Foundation</i>	Spring 2017, Fall 2019 Spring 2020, Fall 2020
Certificate of Appreciation (OUTTA) <i>Dean, college of Engineering, Seoul National University</i>	Jun. 2021
Scholarship to Academic Excellence <i>Seoul National University</i>	Spring 2015, Fall 2015 Spring 2016, Fall 2016

Graduate courses

Specialization: Non-convex Optimization & Reinforcement Learning

Theoretical statistics I, II, Probability Theory I, II

Convex Optimization (convex optimization, robust optimization)

Mathematical Programming II (Advanced optimization theory, non-convex optimization)

Advanced control system I (canonical state-space representation forms, Lyapunov stability, LQR control)

Experiential advanced control design I, II (model predictive control, kalman filter)

Linear System, Nonlinear System

Technical Skills

Languages: Python (Advanced), MatLab (Advanced), C++

Software library, platform : Pytorch (Advanced), Tensorflow. Gurobi (Advanced), CPLEX (Advanced)